Literature search report – Example project

scitech-librarian 3.2

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Generated by scitech-librarian 3.2 (report level: simple). Every number below is reproducible from the archived research directory `lit`.

Item	Value
Search dates	2026-08-28 to 2026-08-28
Project	Example project
Description	the four example blocks of queries.example.json, searched twice, plus a reference list from a colleague; CC0 sources only
Sources	2 automated run(s), 1 manual source(s)
Blocks	NOV, MLP, PSC, QC
Backends / sources	openalex, arxiv, inspire, manual:reference-list
Mode	full fetch, up to 300 records per block and backend
Non-curated venue filter	on
Open-access lookup	not run
Journal metrics	on file for 103 journals
Filters	sources=all

Sources

Every search that feeds this report, oldest first. 'New here' counts unique records that no earlier source had found – what each search added.

Source	Kind	Date	Method	Records	New here	Label / origin
20260828T091142	run	2026-08-28 09:11	database	5	5	first pass (OpenAlex only, NOV + MLP)
20260828T160426	run	2026-08-28 16:04	database	1286	1221	full run, CC0 databases (OpenAlex, arXiv, INSPIRE)
reference-list	manual	2026-08-28 16:07	citation	8	2	reference list of a review article

Search strategy

Each block is one structural query – a conjunction of synonym groups, (a OR b) AND (c OR d) – rendered into every backend's native grammar. The strings below are exactly what was sent (PRISMA-S item 8), from the most recent run of each block.

Block NOV: EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps

Purpose: novelty-check pattern – a SMALL number is the good outcome; read every hit by hand before claiming a gap

```
(origami OR kirigami) AND (acoustic metamaterial OR phononic crystal) AND (topological pumping OR
Thouless pump OR edge state)
```

Backend	Query string sent
openalex	(origami OR kirigami) AND ("acoustic metamaterial" OR "phononic crystal") AND ("topological pumping" OR "Thouless pump" OR "edge state")
arxiv	(all:"origami" OR all:"kirigami") AND (all:"acoustic metamaterial" OR all:"phononic crystal")
inspire	("origami" or "kirigami") and ("acoustic metamaterial" or "phononic crystal") and ("topological pumping" or "Thouless pump" or "edge state")

Block MLP: Machine-learned interatomic potentials for amorphous materials

Purpose: a narrower intersection – expect hundreds; good for testing record fetch and RIS export

```
(machine learning potential OR machine-learned interatomic potential OR neural network potential) AND
(amorphous OR glass OR disordered solid) AND (molecular dynamics OR structure prediction)
```

arXiv receives groups [0, 1] only (nested-boolean limitation).

Backend	Query string sent
openalex	("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND (amorphous OR glass OR "disordered solid") AND ("molecular dynamics" OR "structure prediction")
arxiv	(all:"machine learning potential" OR all:"machine-learned interatomic potential" OR all:"neural network potential") AND (all:"amorphous" OR all:"glass" OR all:"disordered solid")
inspire	("machine learning potential" or "machine-learned interatomic potential" or "neural network potential") and ("amorphous" or "glass" or "disordered solid") and ("molecular dynamics" or "structure prediction")

Block PSC: Perovskite solar cell degradation under humidity

Purpose: grounding block – expect thousands of hits; use it to sanity-check that every backend is reachable and the counts are plausible

```
(perovskite solar cell OR halide perovskite photovoltaic) AND (degradation OR stability OR encapsulation)
 AND (humidity OR moisture OR water ingress)
```

Backend	Query string sent
openalex	("perovskite solar cell" OR "halide perovskite photovoltaic") AND (degradation OR stability OR encapsulation) AND (humidity OR moisture OR "water ingress")
arxiv	(all:"perovskite solar cell" OR all:"halide perovskite photovoltaic") AND (all:"degradation" OR all:"stability" OR all:"encapsulation")
inspire	("perovskite solar cell" or "halide perovskite photovoltaic") and ("degradation" or "stability" or "encapsulation") and ("humidity" or "moisture" or "water ingress")

Block QC: Quasicrystal photonics (arXiv-tuned example)

Purpose: demonstrates arxiv_groups: arXiv chokes on deeply nested booleans, so name the two groups it should get

```
(quasicrystal OR quasiperiodic lattice) AND (photonic OR waveguide OR optical lattice) AND (localization
OR critical states OR fractal spectrum)
```

arXiv receives groups [0, 2] only (nested-boolean limitation).

Backend	Query string sent
openalex	(quasicrystal OR "quasiperiodic lattice") AND (photonic OR waveguide OR "optical lattice") AND (localization OR "critical states" OR "fractal spectrum")
arxiv	(all:"quasicrystal" OR all:"quasiperiodic lattice") AND (all:"localization" OR all:"critical states" OR all:"fractal spectrum")
inspire	("quasicrystal" or "quasiperiodic lattice") and ("photonic" or "waveguide" or "optical lattice") and ("localization" or "critical states" or "fractal spectrum")

Results summary

Block	openalex	arxiv	inspire	manual:reference-list	Identified	Retrieved	Unique
NOV	0	0	0	-	0	0	0
MLP	477	138	0	8	623	357	315
PSC	4569	175	0	-	4,744	475	474
QC	154	414	17	-	585	467	439
Total	5,200	727	17	8	5,952	1299	1228

Identified = database hit counts (not comparable across backends: proximity operators are dropped and stemming differs); summed over runs, and for manual sources the number of records ingested. Retrieved = records actually downloaded after the venue filter, capped by `--limit`. Unique = after DOI/title deduplication across all sources.

Timeline

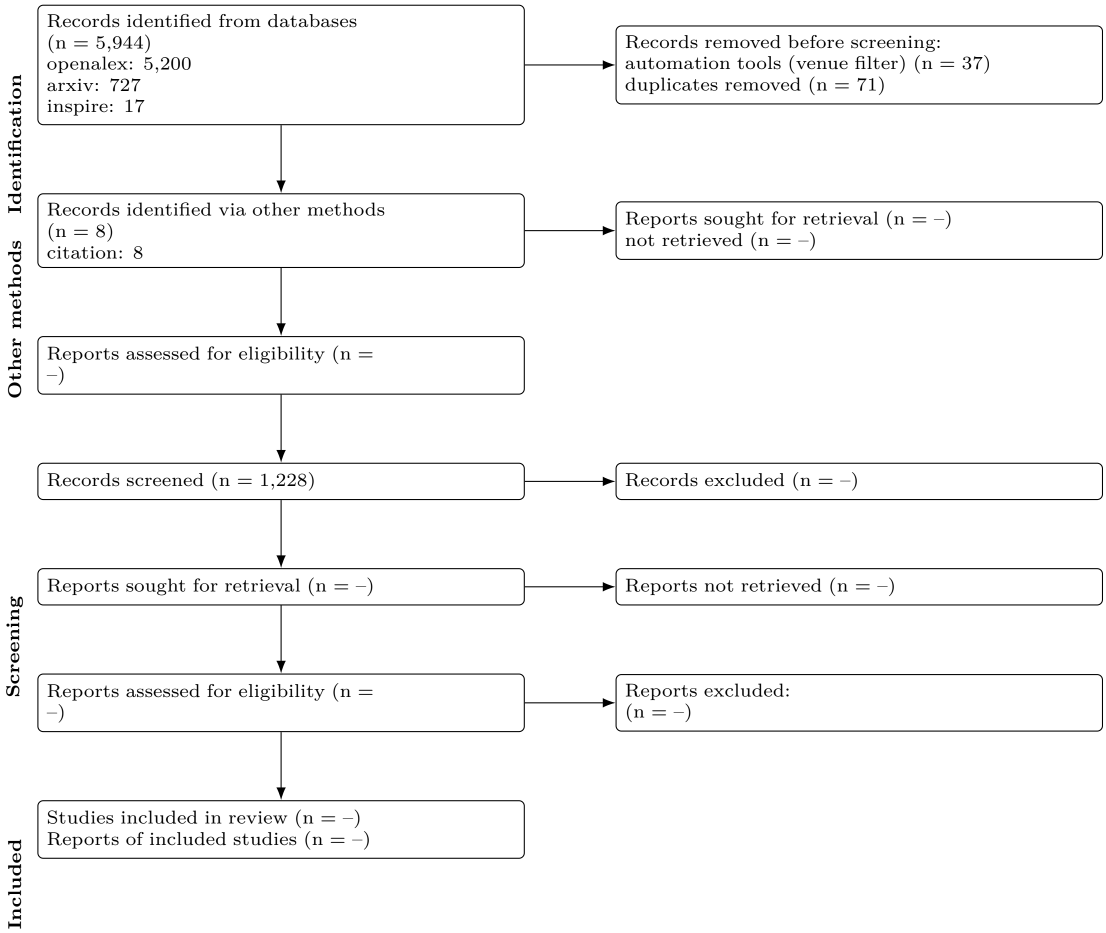
Per-block hit totals in each automated run (sum over backends), oldest first; drift shows how the indexes – or the queries – changed.

Block	20260828T091142	20260828T160426
NOV	0	0
MLP	239	376
PSC	-	4,744
QC	-	585

When records entered the project

Month	Records first seen
2026-08	1228

PRISMA 2020 flow



Stage	n
Records identified from databases	5,944
openalex	5,200
arxiv	727
inspire	17
Records identified via other methods	8
citation	8
Records retrieved (downloaded / ingested)	1,336
Removed before screening: automation (non-curated venues)	37
Removed before screening: duplicates	71
Records to screen (unique)	1,228
Records screened	1,228 (assumed = unique)
Records excluded at screening	–
Reports sought for retrieval	–
Reports not retrieved	–
Reports assessed for eligibility	–

Stage	n
Other methods: reports sought	–
Other methods: reports not retrieved	–
Other methods: reports assessed	–
Studies included	–
Reports of included studies	–

Automation stages are computed from the data; '–' marks manual stages not yet recorded in screening.json. Note that 'identified' counts hits reported by each database while 'retrieved' is what was downloaded within `--limit`, so the two differ on large blocks.

PRISMA-S search-reporting checklist

Item	Requirement	This search
1	Database name	openalex, arxiv, inspire (documented public APIs)
2	Multi-database searching	3 databases, one structural query per block rendered into each native grammar; see Search strategy
3	Study registries	not applicable
4	Online resources and browsing	none recorded
5	Citation searching	reference-list
6	Contacts	none recorded
7	Other methods	none
8	Full search strategies	reported verbatim per backend under Search strategy; archived in queries.json of each run
9	Limits and restrictions	record download capped at 300 per block and backend, most-cited first; no date, language or document-type limits applied; report filters: sources=all
10	Search filters	venue filter on: records from non-curated repositories (Zenodo, Figshare, SSRN...) removed
11	Prior work	none
12	Updates	2 run(s) combined; see Timeline
13	Dates of searches	searched on 2026-08-28 to 2026-08-28
14	Peer review	none
15	Total records	5,944 identified from databases, 8 via other methods; 1,336 retrieved; 1,228 unique
16	Deduplication	exact DOI match, else first 90 characters of the lower-cased title; 71 duplicates removed

Top 10 records per block

Deduplicated across sources, sorted by cited. The complete set is in all_records.csv / .ris of each run.

Block MLP (315 unique)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Atom-centered symmetry functions for constructing high-dimensional neural network potentials	Jörg Behler	2011	The Journal of Chemical Physics	1644	10.1063/1.3553717	
Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics	Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.	2018	The Journal of Physical Chemistry Letters	269	10.1021/acs.jpclett.8b00902	
Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide	Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al.	2020	npj Computational Materials	185	10.1038/s41524-020-00367-7	
Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide	Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al.	2019	Cambridge University Engineering Department Publications Database	184	10.17863/cam.55408	(2026)
Growth Mechanism and Origin of High \$sp^3\$ Content in Tetrahedral Amorphous Carbon	Miguel A. Caro; Volker L. Deringer; Jari Koskinen et al.	2018	Phys. Rev. Lett. 120, 166101 (2018)	178	10.1103/PhysRevLett.120.166101	
Constructing first-principles phase diagrams of amorphous LixSi using machine-learning-assisted sampling with an evolutionary algorithm	Nongnuch Artrith; Alexander Urban; Gerbrand Ceder	2018	The Journal of Chemical Physics 148 (24), 241711 2018	155	10.1063/1.5017661	
Study of Li atom diffusion in amorphous Li3PO4 with neural network potential	Wenwen Li; Yasunobu Ando; Emi Minamitani et al.	2017	The Journal of Chemical Physics	154	10.1063/1.4997242	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Impact of the Local Environment on Li Ion Transport in Inorganic Components of Solid Electrolyte Interphases	Taiping Hu; Jianxin Tian; Fuzhi Dai et al.	2022	Journal of the American Chemical Society	107	10.1021/jacs.2c11521	
Thermal conductivity modeling using machine learning potentials: application to crystalline and amorphous silicon	Xin Qian; Shiyu Peng; Xiaobo Li et al.	2019	Materials Today Physics	103	10.1016/j.mtphys.2019.100140	
A unified deep neural network potential capable of predicting thermal conductivity of silicon in different phases	Ruiyang Li; Eungkyu Lee; Tengfei Luo	2020	Materials Today Physics	102	10.1016/j.mtphys.2020.100181	

Block PSC (474 unique)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance	Michael Saliba; Taisuke Matsui; Konrad Domanski et al.	2016	Science	3691	10.1126/science.aah5557	
High-efficiency two-dimensional Ruddlesden–Popper perovskite solar cells	Hsinhan Tsai; Wanyi Nie; Jean-Christophe Blancon et al.	2016	Nature	3361	10.1038/nature18306	
Efficient, stable and scalable perovskite solar cells using poly(3-hexylthiophene)	Eui Hyuk Jung; Nam Joong Jeon; Eun Young Park et al.	2019	Nature	2318	10.1038/s41586-019-1036-3	
A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability	Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.	2014	Angewandte Chemie International Edition	2006	10.1002/anie.201406466	15.006 (2026)
Review of recent progress in chemical stability of perovskite solar cells	Guangda Niu; Xudong Guo; Liduo Wang	2014	Journal of Materials Chemistry A	1945	10.1039/c4ta04994b	
Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics	Caleb C. Boyd; Rongrong Cheacharoen; Tomas Leijtens et al.	2018	Chemical Reviews	1767	10.1021/acs.chemrev.8b00336	53.886 (2026)
Formamidinium and Cesium Hybridization for Photo- and Moisture-Stable Perovskite Solar Cell	Jin-Wook Lee; Deok-Hwan Kim; Hui-Seon Kim et al.	2015	Advanced Energy Materials	1661	10.1002/aenm.201501310	15.613 (2026)
23.6%-efficient monolithic perovskite/silicon tandem solar cells with improved stability	Kevin A. Bush; Axel F. Palmstrom; Zhengshan J. Yu et al.	2017	Nature Energy	1533	10.1038/nenergy.2017.9	
Investigation of CH 3 NH 3 PbI 3 Degradation Rates and Mechanisms in Controlled Humidity Environments Using in Situ Techniques	Jinli Yang; Braden D. Siempelkamp; Dianyi Liu et al.	2015	ACS Nano	1342	10.1021/nn506864k	16.894 (2026)
Carbon Nanotube/Polymer Composites as a Highly Stable Hole Collection Layer in Perovskite Solar Cells	Severin N. Habisreutinger; Tomas Leijtens; Giles E. Eperon et al.	2014	Nano Letters	1198	10.1021/nl501982b	

Block QC (439 unique)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Hyperuniform states of matter	Salvatore Torquato	2018	Physics Reports	485	10.1016/j.physrep.2018.03.001	
Localization and delocalization of light in photonic moire lattices	Peng Wang; Yuanlin Zheng; Xianfeng Chen et al.	2020	Nature 577, 42-46(2020)	482	10.1038/s41586-019-1851-6	
Delocalization of a disordered bosonic system by repulsive interactions	B. Deissler; Matteo Zaccanti; G. Roati et al.	2010	Nature Physics	273	10.1038/nphys1635	
Topological triple phase transition in non-Hermitian Floquet quasicrystals	Sebastian Weidemann; Mark Kremer; Stefano Longhi et al.	2022	Nature	225	10.1038/s41586-021-04253-0	
Disorder-Enhanced Transport in Photonic Quasicrystals	Liad Levi; Mikael C. Rechtsman; Barak Freedman et al.	2011	Science	205	10.1126/science.1202977	
Generalized Aubry-André self-duality and mobility edges in non-Hermitian quasiperiodic lattices	Tong Liu; Hao Guo; Yong Pu et al.	2020	Physical review. B./Physical review. B	148	10.1103/physrevb.102.024205	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Simulating non-Hermitian quasicrystals with single-photon quantum walks	Quan Lin; Tianyu Li; Lei Xiao et al.	2021	Phys. Rev. Lett. 129, 113601 (2022)	136	10.1103/PhysRevLett.129.113601	
Localization and delocalization of light in photonic moiré lattices	Peng Wang; Yuanlin Zheng; Xianfeng Chen et al.	2020	LA Referencia (Red Federada de Repositorios Institucionales de Publicaciones Científicas)	121	(link)	
Bose-Einstein Condensates in Optical Quasicrystal Lattices	Laurent Sanchez-Palencia; Luis Santos	2005	Physical Review A 72 (2005) 053607	117	10.1103/PhysRevA.72.053607	
Observing Localization in a 2D Quasicrystalline Optical Lattice	Matteo Sbroscia; Konrad Viebahn; Edward Carter et al.	2020	Physical Review Letters	105	10.1103/physrevlett.125.200604	

Suggestions

- Block NOV: zero hits on every backend. Either the intersection is genuinely empty (a finding – check the synonyms first) or one group is too narrow; try dropping one group and rerunning.
- Block PSC: openalex 4,569 hits – a generic term is probably driving this; tighten a group or add a more specific one before reading.
- Block PSC: counts differ >20x across backends (175 to 4,569); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- Block QC: counts differ >20x across backends (17 to 414); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- No citation-grade backend (Scopus, Web of Science, NASA ADS) was in this search; add ADS (free token) or Scopus before quoting counts.
- Open-access status was not looked up; rerun with `--pdfs` (optionally `--pdf-blocks`) to collect legal OA PDF links via Unpaywall.
- The PRISMA flow's manual stages are empty: fill in screening.json (screened / excluded / assessed / included) and rerun report.py to complete the diagram.